Plane mirrors

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use the reflection law and locate a virtual image.

SUCCESS CRITERIA - I can...

1. Measure angles from the normal.
2. Use equal incidence and reflection angles.
3. Place the image the same perpendicular distance behind the mirror.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “normal line” and “reflection” to explain your model or result. Sentence starters are optional.

normal line: a reference line perpendicular to a surface

reflection: a wave returning from a boundary

virtual image: an image at an apparent origin of rays, not their meeting point

01 Warm-up: bring back an earlier idea

A wall is 50 m away. An echo returns after 0.5 s. A student divides one-way distance by this time.
Find the correct speed and explain the student’s error.

Word help: echo = a sound received after reflection

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: ① ② ③

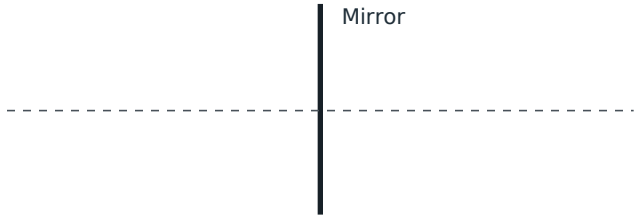
02 Worked example: explain the model

Reference: $\theta_{incident} = \theta_{reflected}$
image distance behind = object distance in front.
Plane mirror; measure angles from the normal. The image is virtual.

Worked reasoning: An object 2 m in front of an ideal plane mirror has a virtual image 2 m behind it. The object-image separation is 4 m; reflected rays appear to come from the image.

Your turn: Draw the mirror, object, image and backward extensions of two reflected rays.

Sentence starter: The … in my model represents … . Its label shows … .



Add object / image or incident / reflected rays.

Marzano check - circle: ① ② ③

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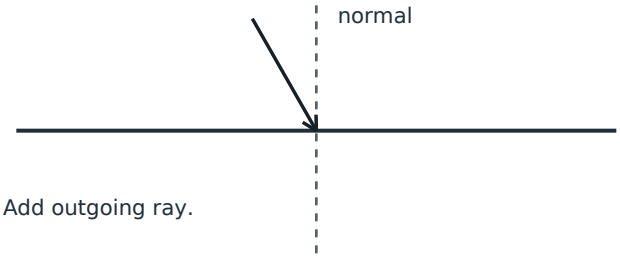
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Draw the normal at the mirror. Complete a reflected ray for an incident ray 30° from normal.

Sentence starter: The … in my model represents … . Its label shows … .

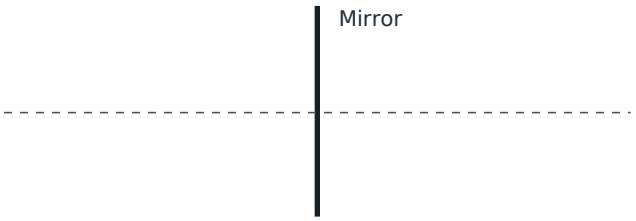


Marzano check - circle: ① ② ③

04 Apply the physics

A small lamp is 2 m in front of a plane mirror.
Mark the lamp and its virtual image on the model.
Find image distance behind the mirror and the full lamp-to-image separation.

Sentence starter: My design shows … . It meets the requirement by … .



Marzano check - circle: ① ② ③

05 Explain your reasoning

Why is the mirror image called virtual rather than a real object behind the glass?

Sentence starter: My claim is … . The diagram/data show …, so … .

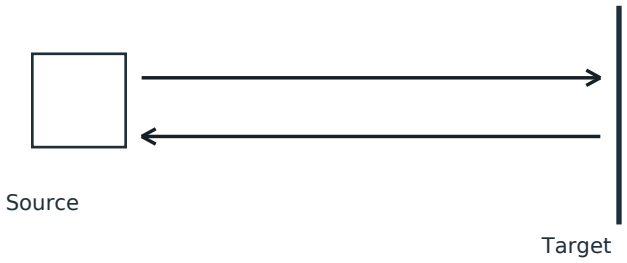
Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A reflected signal returns after 4 s.
Name the extra quantity needed to calculate distance and explain why this is a round trip.
Label the outgoing and returning paths.

Word help: signal = a disturbance that carries information

Sentence starter: My claim is … . The diagram/data show …, so … .



Marzano check - circle: ① ② ③

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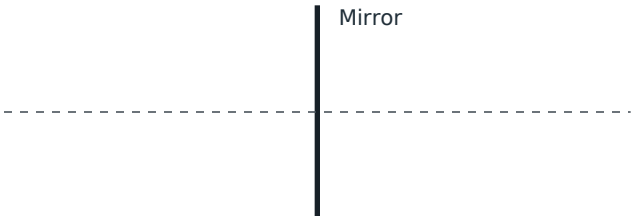
Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

Incidence angle 45° from normal.
Find reflection angle.

Sentence starter: I found ... [unit]. This value means ... in this situation.



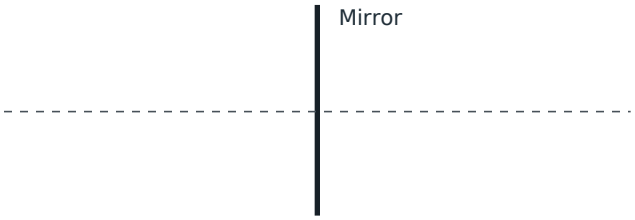
Add object / image or incident / reflected rays.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Move the object 0.5 m closer to the mirror.
Find the new image distance and object-image separation.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Add object / image or incident / reflected rays.

Marzano check - circle: 1 2 3